

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2016

PLEASE ENSURE THAT YOU HAVE SUBMITTED ONE DOWNLOADED
ONLINE REGISTRATION FORM BEFORE AND AFTER THE EXAMINATION

TIME: 1Hour 30 Mins.

Answer all questions

1. If $y = (1-2x)^3$, find the value of $\frac{dy}{dx}$ at $x = -1$. A. 57 B. 27 C. -6 D. -54

2. In how many ways can 6 colored chalks be arranged if 2 are of the same colour? A. 608 B. 120 C. 240 D. 360

3. How many possible ways are there of seating seven people P, Q, R, S, T, U, V at a circular table? A. 360 B. 720 C. 2520 D. 5040

4. What is the probability that an integer x , ($1 < x < 20$) chosen at random is divided by both 2 and 3? A. $\frac{1}{20}$ B. $\frac{1}{3}$ C. $\frac{3}{20}$ D. $\frac{7}{10}$

5. Find $\lim_{x \rightarrow 0} \frac{4516 - P_x}{P_x} = 3056$. A. 117 B. 1147 C. 116 D. 1157

6. Find the sum to infinity of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$. A. $\frac{1}{6}$ B. $\frac{3}{4}$ C. $\frac{2}{3}$ D. $\frac{1}{3}$

7. y is inversely proportional to x and $y = 4$ when $x = 4$. Find x when $y = 1$.

8. Given that $Y^4 = 16$, find x . A. 2 B. 3 C. 4 D. 6

9. $\int \frac{1}{x^2} dx =$ _____

10. Find the derivative of $(2+3x)^2(1-x)$ with respect to x . A. $6x-1$ B. $1-6x$ C. 6 D. 3

11. Find the derivative of the function of $y = 2x^2(2x-1)$ at $x = 1$.

12. Find the mean deviation of 1, 2, 3, and 4. A. 1.0 B. 2 C. 3 D. 4

13. In how many ways can 2 students be selected from a group of 5 students in a debating competition? A. 10 ways B. 15 ways C. 20 ways D. 25 ways

14. If the interest on N150.00 for 2% yrs is N4.50. Find the interest on N250.00 for 6 months at the same rate. A. N1.50 B. N7.50 C. N15.00 D. N18.00

15. What are the integral values of x which satisfy $A = -2, 1, 0, -1$?

B.-1,0.1.2C.-1,0,1D.0.1.2

- 16.If the 29th tem of an AP is twice the third tem and the sum of the first four term is 42.Find the cammon difference.A.6 B.3 C.2 D.1
- 17.Find the sum of the first 20 term of the series 8,12,1696
A. 1400 B. 1500 C. 1040 D. 960
- 18.Given the series 24.6 find the 4 term of the GP A.32 B.68
C. 70 D. 64
- 19.The sum of the interior angles of a regular polygon is 1800.Calculate the size of the exterior angle of the polygon.A.30°B.24°C.15°0.12°
- 20.Find theremainder when $3x^3+5x^2-11x+4$ is divided by $x+3$
A.4B.1C-1D.-4
- 21.The first and lat term of an AP is o and 42.If the sum of the Progression is 192,the number of the term in the APis
A. 68. 7 C. 80. 5
22. ...
- 23.The real number x which satisfy the inequality $\frac{1}{3}(x\pm 1)-1 > \frac{1}{5}(x+4)$.A.x<11B.x=1C.x>1D.x=11
- 24.If B and aaretheroots of $5x^2-x=10$,a+BisA.B.1/5
C. 10D. 2
- 25.What isthe 4th angle for the quadrilateral whose dther three angles are 580, 1170, 122°A. 30 B. 60° C. 63° 0. 360°

PHYSICS TYPE A

- 26.In which of the following material would sound travel least.A.water B.oil C metal D.gas
- 27.Suppose the direction of the refracted ray is required for the light incident in air at 50°,n=1.33.Find the angle of refraction A.45°B.50°C.70°D.35°
- 28.The abiity of the eye lens to focus point at different distance on to the retina is A.Binocular B.Accommodation C.Hypermotropia D.Colourization
- 29.When light is refrated which colour is the least refracted?A.Red B.Blue C.Violet D.indigo
- 30.The frequency F of the fundamental note from plucked wire

$$B \cdot \frac{1}{21} \cdot \frac{m}{c}$$

- 31.Suppose the potential difference of 2.4V at a current of 0.2A.Find the resistance A.14?B.12?C,16?D.20?
- 32.Which is correct for dimension of acceleration A.MLT²
B.LT³C.LT²D.LT
- 33.Suppoe a car of mass 1000kg is accelerating at 2m/s² then the force F acting on it is A.20000N B.2000N
C. 40000N D. 4500N
- 34.Whch of the same quantities has same unit as Joules A. Newton X Foce B.Force X acceleration C.Newton X meter D.Watt XForce
- 35.If the pie driver has a weight of 660N and is raied 20m, then energy of stationary pile is A.13200JB.134003
C. 14200JD. 15200J
- 36.The VR is independent of A Quaity B.pulley C. friction D.frequency
- 37.The process whereby the mdecules of dfferent substance move more randomly is called A.surface tension B.Capillarity C.diffusion D.osmosis
- 38.The process whereby a liquid spontanecusly changes into vapour is caled A.evaporation B.regulation C. boiling D.sublimation

- 39.The pitch of sound note depend on A.timber B.Harmonics C.Frequency D.Quality.
- 40.Satelite communication network makes use of A.Infra-red ray B.Sound wave C.Radio wave D.x-ray wave
- 41.If two inductors of inductance 3H and 6H are arrange in series,the total inductance His A. 18.0H B.9.0H C.2.0H
D.5.0H
- 42.The charge carriers in gases are A.ions only B.electrons only C.electrons and ions D.protons only
- 43.The mode of heat transfer that do not require a material medium.A.conduction.8.convection C.radiation D. propagation
- 44.The unit for energy is A.Joule B.Ampere C Watts D. Pascal
- 45.The magnetic field is produce at what ange A.0°D.30°
C. 180° D. 90°
- 46.The frequency of a stationary wave of 100Hz has a wavelength of 2m.Find the velocity of waves A.500m/s B. 200m/sC. 300m/s D. 40m/s
- 47.A battery of e.m.f 40V and internal resistance 5Q is connected to a resistance of 150.Calculate the terminal potential difference A.40W B.20V C.30V D.75V
- 48.A car moving round a ciraular racing track takes 120s to do a lap of 8km.what is the speed in km/h?A.250km/h B. 240km/hC.300kni/h D.280km/h
- 49.Filament is commorly used at home because of its A.high freezing point B.low melting point C.low boiling point D. high melting point.
- 50.Secondary cells are better referred to as A.cel B. accumulator C.battery D.charger

CHEMISTRY TYPE A

- 51.When sodium react with water,the resulting solution is A.weakly acidic B.neutral C.acidic D.alkaline
- 52.The gas that gives brown colouration in brown ring test is A.CO B.NO C.CO₂D.NO₂
- 53.Whichof the following polymeris suitable for packaging electrical insulator?A.polystyrene B. polyethylene C.polyamide D.polycarbonate
- 54.The boling of fat and aqueous caustic soda is referred to as
A.Acidification B.Hydrolysis C.Specification D. Etherification
- 55.Hydrogen is released when diute hydrochloric acid react with A.Ag B.Au C.Cu D.Na
- 56.Which of the following chloride would exhibit the least ionic character?A.LiCl B.MgCl₂C.CaCl₂D.AlCl₃
- 57.When H₂Sis passed into a solution of iron (i)chloride,the solution turns A.Brown B.Palegreen C.Colourless D.Red
- 58.The property which makes alohol sdluble in water is the A.cooling B.wetting C.covalent nature D.bonding
- 59.Deliquescent substances are used far A.cooling B.boiling C.drying D.melting
- 60.What mass ofwater is produced when 8.0g of hydrogen react with excess oxygen?-A.36.0g B.8.0g
C. 720g D. 16. 0g (H=1. 0=16)
- 61.Cancerous are cured by exposure to A.y-ray B,aray C.P-ray D.x-ray
- 62.Anoxide XD₂has vapour density of 32.What is the atomic

- mass of X? A.32 B.20 C.14 D.72.
63. Petroleum oil are separated by A. Crystallization B. Fractional Distillation C. Centrifugation D. Decantation.
64. How many moles of CaCO_3 will be required to produce 5.6g of CaO ? (Ca 40, C=12, O=16) A. 0.20mol B. 0.1mole C. 0.25mole D. 0.40mole
65. The compound represented thus C_2H_2 is A. Alkane B. Ethyne C. Ethene D. Ethane
66. The shape of d-orbital is A. Spherical B. orthogonal C. polygonal D. dumbbell
67. The general formula for the alkanone is A. R_2CO B. ROH C. RCOOR D. RCHO .
68. Which of the following metal burns with a brick-red flame A. Pb B. Ca C. Na D. Mg
69. A burning candle produces water and A. carbon (ii) oxide B. carbon (iv) oxide C. oxygen D. hydrogen
70. Proteins in acid solution undergo A. Polymerization B. Substitution C. Fermentation D. hydrolysis
71. Ethanoic acid is A. Tribasic B. Unionizable C. Monobasic D. Diabasic
72. What quantity will be deposited by a current of 2A at a time of 4s A. 9C B. 10G C. 8C D. 12C
73. Phosphorus is stored under water to prevent it from A. dehydrating B. becoming inert C. catching fire D. smelling
74. What is the product of bombarding ^{12}C nucleus with

6

75. The element with the electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^2$ is A. C B. Si C. N D. C₆₀

ENGLISH LANGUAGE TYPE A

76. The principal was not aware the visitors on their arrival A. to have met B. that he should have met C. that he met D. to meet the visitors on their arrival
77. John was glad A. were the exams over B. the exams were over C. the exams to be over D. at the exams which over
78. Patrick was afraid A. whether he had failed B. to have failed C. he has failed D. he had failed
79. Patrick was wrong A. he contradicted the teacher B. to contradict the teacher C. contradicted the teacher D. to be contradicted the teacher
80. Doing snippets of work each day is not anyone A. very satisfactory B. really satisfactory to very C. really very satisfactory to D. very really satisfactory to
81. At last the aircraft was able to take off. It for two hours by a fault in the electrical system A. has been delayed B. delayed C. had been delayed D. is delayed

82. Mr. Momoh hard for many years before he got promotion A. had been worked B. had been working C. has worked D. has been working
83. I am sorry ... to the meeting last night A. not come B. I didn't come C. to come D. I hadn't come
84. John felt sorry the poor people who hadn't enough to eat A. for B. because C. of D. that
85. Mary's was unable the work in time A. she completed B. she couldn't complete C. complete D. not to complete
86. Mr. Olayinka was delighted A. you to come B. that you come C. you not to come D. did you come
87. Food prices a lot since last year A. have gone up B. had gone up C. went up D. go up
88. The Romans once most of Europe and North Africa for many years A. have ruled B. had been ruling C. rule D. ruled
89. Eze has not heard anything of his sister Ada since she to the United Kingdom A. has gone B. had gone C. had been going D. went
90. Everyone looked to Peter to give a lead A. to B. out C. up D. at
91. Everyone looked Peter as their leader A. at B. on C. for D. up
92. Mary would never have finished her homework if Patrick hadn't helped her A. in B. with C. out D. off
93. John spent hours looking his dictionary he couldn't find it A. at B. in C. for D. to
94. Everyone looked to Peter as a great leader A. up B. down C. on D. for
95. At the beginning of the period the geography master fired a series of questions about South East Asia A. up B. on C. away D. off

From each numbered list choose the word or expression which will complete the corresponding sentence correctly.

96. The answer to this sum is wrong, you a mistake A. must make B. must have made C. can have made D. may have made
97. "Hurry up, it's time to go". Surely, it A. can't be B. must be C. must's be D. can be
98. A. Bridget: I'll have to go now Ada, Ada: surely you go yet A. mustn't B. need to C. don't need to D. have got to
99. When you are ready, you go A. may B. can't C. ought not to D. would
100. I wonder where C. K. is he by now A. ought to arrive B. ought to have arrived C. must arrive D. can't have arrived

SOLUTION TO LUTH SCHOOL OF NURSING 2016 EXAM

1. C	2. D	3. A	4. C	5. C	6. B	7. B	8. B	9. C	10. A
11. C	12. A	13. A	14. A	15. D	16. 8	17. C	18. "	19. A	20. 8
21. C	22. C	23. C	24. B	25. C	26. 8	27. D	28. B	29. C	30. D
31. B	32. C	33. B	34. C	35. A	36. C	37. C	38. A	39. C	40. C
41. C	42. B	43. B	44. A	45. D	46. B	47. C	48. B	49. D	50. B
51. D	52. D	53. A	54. C	55. A	56. D	57. D	58.	59. C	60. A
61. D	62. A	63. D	64. B	65. B	66. D	67. A	68. B	69. B	70. A
71. C	72. C	73. C	74. B	75. B	76. C	77. 8	78. C	79. 8	80. C
81. 8	82. C	83. B	84. A	85. C	86. 8	87. C	88. D	89. D	90. C
91. 8	92. C	93. C	94. A	95. B	96. B	97. C	98. C	99. A	100. B

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TEST 2015

TIME:1 HOUR

INSTRUCTIONS TO CANDIDATES: Please ensure that you have submitted one downloaded on-line registration form before and after the examination. Calculators, cell phones and other electronics devices are not allowed. Any form of examination malpractice automatically disqualifies the candidate. Candidates may use logarithm tables. Answer all questions TIME ALLOWED: 1hr 30mm

Shade the answer sheet as appropriate: with HB pencil only

PHYSICS

- The slope of a straight line displacement time graph indicates
A. distance travelled B. uniform velocity C. uniform acceleration D. instant acceleration E. uniform speed.
- A ball of mass 0.5kg moving at 10m/s collides with another ball of equal mass at rest. If the two balls move off together after the impact, calculate their common velocity A. 0.2m/s B. 0.5m/s C. 10m/s D. 3m/s
- How much heat is given out when a piece of iron mass 50g and specific heat capacity 460/kg/K cools from 85°C to 25°C?
A. 1.38×10^6 J B. 5.3×10^2 C. 1.98×10^2 D. 1.38×10^3
E. 1.27×10^3 J
- Which of the following is not a suitable method of reducing loss of heat from a piece of hot iron? A. wrapping it in cotton wool B. painting it black C. placing it in a vacuum D. placing it in a rubber support E. keeping it in a closed wooden box
- A bat emits a sound wave at a speed of 1650.00m/s and receives the echoes 0.15s later. Calculate the distance of the bat from the reflector A. 8.75m B. 16.6Cm C. 87.75m D. 123.75m E. 330.00m
- Which of the following is/are characteristics of sound? i. pitch ii. Loudness iii. Quality iv. Noise A. i only B. ii only C. i & ii only D. i, ii & iii only E. i, ii, iii and iv
- An image which can be formed on a screen is said to be
A. virtual B. blurred C. inverted D. erect E. real
- A ray of light is incident at an angle of 30° on a glass prism of refractive index 1.5. Calculate the angle through which the ray is minimally deviated in the prism. (the medium surrounding the prism is air) A. 10.5° B. 5.5° C. 21.1° D. 38.9° E. 40.5°
- At which of the following distances from the lens should a slide be placed in a slide greater than f but less than 2f
D. equal to f E. equal to 2f
- What of the camera corresponds to the iris of the eye?
A. shutter B. film C. lens D. diaphragm E. focusing ring
- Which of the following is not a mechanical wave? A. wave propagated in stretched string B. waves in closed pipes C. radio waves D. water waves E. sound waves
- A catapult is used to project a stone. Which of the following energy conversions takes place as the stone is released? A. The kinetic energy of the stone is converted into gravitational potential energy B. The gravitational potential energy of the catapult is converted into the kinetic energy of the stone C. The elastic potential energy of the catapult is converted into elastic potential energy D. The elastic potential energy of the catapult is converted into the

gravitational potential energy of the stone

- Ball is thrown vertically upwards from the ground with an initial velocity of 50m/s. What is the total time spent by the ball in air? ($g = 10\text{m/s}^2$) A. 2.5s B. 5.0s C. 10.0s D. 15.0s E. 20.0s
- Which of the following correctly gives the relationship between linear speed V and angular speed ω of a body moving uniformly in a circle of radius r? A. $v = r\omega$ B. $v = \omega^2 r$ C. $v = \omega r^2$ D. $v^2 = \omega r$ E. $v = \omega r$
- Which of the following is used to determine the relative density of the acid in a car battery? A. hydrometer B. hygrometer C. manometer D. hydrometer E. spectrometer
- A block of material of volume 20cm³ and density 2.3g/cm³ is suspended from a spring balance with half the volume of the block immersed in water. What is the reading of the spring balance? (Density of water = 1.0g/cm³) A. 8g B. 25g C. 30g D. 40g E. 50g
- A piece of cork which is floating on water is acted upon by the forces of A. weight and viscosity B. weight and upthrust C. upthrust and viscosity D. weight only E. upthrust only
- Which of the following is derived unit? A. meter B. coulomb C. kilogram D. second E. ampere
- An engine raises 100kg of water through a height of 60m in 20s. What is the power of the engine? ($g = 10\text{m/s}^2$) A. 120.00W B. 3000.00W C. 333.00W D. 300.00W E. 30.00W
- If the temperature of a small quantity of water in a closed container is gradually increased from 0°C to 40°C then the density of water within its range A. increases for a while and then decreases B. decreases for a while and then increases C. increases gradually D. decreases gradually E. remains the same

MATHEMATICS

- Simplify: $125\sqrt[3]{4^2 \times 10^3}$ A. 350 B. 35 C. $\frac{1}{3}$ D. $\frac{1}{30}$ E. 0
- If $3^{2x} = 27$, what is x? A. 1 B. 1.5 C. 4.5 D. 18 E. 40.5
- Express 0.00562 in standard form. A. 5.62×10^3 B. 5.62 $\times 10^2$ C. 0.562 $\times 10^2$ D. 5.62 $\times 10^5$ E. 5.62 $\times 10^3$
- Given that $\frac{1}{3} \log_{10} P = 1$, find the value of P A. 3 B. $\frac{1}{10}$ C. 10 D. 100 E. 1000
- $25, \text{ mm}^*$ A. $16E3 \times 120$
- The population of a village is 5846. Express this number to three significant figures A. 5850 B. 5846 C. 5840 D. 585 E. 584
- Simplify $((\frac{1}{4})^{\frac{1}{2}})^{\frac{1}{2}}$ A. $\frac{1}{6}$ B. $\frac{1}{10}$ C. $\frac{2}{4}$ D. $\frac{1}{4}$ E. 8
- For what value of y is the expression $(y+2)/(y^2 - 5y - 10)$ undefined? A. y=0 B. y=2 C. y=3 D. y=5 E. y=10
- Simplify $\log_6 + \log_{12} A - 4B - 1C$ A. 0 D. 1 E. 4
- Solve the equation $3a + 10 = a^2$ A. a=5 or a=-5 or

$$a=2Ca=10\text{ora}=0D.a=5\text{ora}=-2E.a=-5\text{ora}=-2$$

31. Evaluate $(101.2)^2 - (100.5)^\circ$ A. 1B. 2. 02 C. 20. 02 D. 141. 19
E. 202

32. Express the product of 0.06 and 0.09 in standard form A.
 5.4×10^2 B. 5.4×10^2 D. 5.4×10^6 D. 5.4×10^2 E. 5.4×10^3

33. Simplify $(36)V^2 \times 5^\circ$ A. $3/4$ B. $1/24$ C. $2/3$ D. $11/2$ E. 2029

34. If $\cos(A)=3$ which of the following angles has a cosine-
k? A. 30° 8. 120° °C. 150° D. 210° E. 330°

35. A ladder 9m long leans against a vertical wall making an
angle of 64° with the horizontal ground. Calculate correctly
the decimal place, how far the foot of the ladder is from
the wall. A. 4.0m B. 5.8m C. 7.1m D. 8.1m E. 18cm

36. If $\log a + 5 \log a - 6 \log a$. What is a? A. 18. B. 6. C. 8. D. G. 16
E. 32

37. If the second and fourth terms of a geometrical progression
are B and 32 respectively, what is the sum of the four
terms? A. 28 B. 40 C. 48 D. 60 E. 68

38. Factorize $35 - 2b - b^2$ A. $(35 - 2b)(b - 1)$ B. $(7 + b)(5 - b)$ C. $(3 + 7)(5 - b)$ D.
 $(35 - b)(3b + 7)$ E. $(7 + b)(5b)$

39.
form. A. 7.4×10^2 B. 7.5×10^1 C. 7.5×10^1 D. 7.5×10^2 E.
 7.5×10^3

40. Factorize the following expression $2x^2 + x - 15$. A. $(2x + 5)$
 $(y - 3)$ B. $(2x - 5)(y + 3)$ C. $(2x - 5)(x - 3)$ D. $(2x - 3)(x + 5)$ E. $(2x + 5)$
 $(x + 3)$

CHEMISTRY:

41. Complete hydrogenation of oils results in the production of
A. soaps B. detergent C. alkalines D. butter F. margarine.

42. A sample of orange juice is suspected to have been
contaminated with a yellow dye. Which of the following
methods can be used to detect the dye? A. decantation B.
Chromatography C. distillation D. filtration E. evaporation

43. The heat accompanying the reaction represented by the
equation $\text{H}_2\text{O(g)} \rightarrow \text{H}_2\text{O(l)}$ is described as the heat of A. Solution B.
neutralization D. sublimation E. activation

44. Which of the following compounds will undergo
addition reactions? A. ethyne B. butane C. pentane D.
Tetrachloromethane E. ethanol

45. How many unpaired electrons are there in an atom
whose atomic number is 8? A. 0 B. 1 C. 2 D. 3

46. A piece of metal object of mass 60 was analysed and found to
contain 20% by weight of magnesium. Calculate the number of
moles of magnesium present (Mg=12) A. 0.5mol
B. 1mol C. 1.5mol D. 2mol E. 2.5mol

47. What is the maximum number of electrons that can
be contained in the d-sub shell A. 6 B. 8 C. 10 D. 12

48. What is the oxidation number of zinc in Zn(OH)_4^{2-} A. -2
B. 4 C. 2 D. -1 E. -4

49. Write the formula of the compound whose name is iron
(ii) Tetraoxosulphate(vi) heptahydrate. A. $\text{Fe}_2\text{SO}_4 \cdot 7\text{H}_2\text{O}$ B.
 $2\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ C. $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ D. $\text{FeSO}_4(\text{H}_2\text{O})_7$

50. Using exponential notation, express the quantity 72.8mg in
terms of its basic S.I. unit A. 7.28×10^3 B. 7.28×10^3 kg
C. 7.28×10^3 g D. 7.28×10^0 kg

51. A given mass at constant temperature occupies 500l at
700mm pressure. What volume will it occupy at 200mm
pressure? A. 1750 B. 1875 C. 1645 D. 5017

52. The expression $P_1 + P_2 +$

ENGLISH LANGUAGE

- This is important.....your last essay. A. on B. with C. by
D. for
- The team has suffered defeat A. before B. with C.
over D. after
- Unemployment has been ... the increase since last year.
A. of B. at C. on D. in
- Cat sleep..... Day and hunt at night A at B. by C. in D.
during
- I don't want anything to interfere..... My project A with
B. on C. over D. by
- The two brothers came in..... The backdoor A. within
B. along C. across D. through
- the terms of agreement with your landlord, you
should not sub-let the house. A. on B. under C. upon D. by
- Nigeria's economy was buoyant..... the 1970s A. through
B. in C. in D. with
- The musician waxed that record died fifty years ago A
who B. whom C. which D. whose
- The author..... book you are reading now, won the
Nobel Prize for literature last year A. which B. that C.
whose D. whom
- The lady..... you were talking about yesterday is here now
A. which B. whom C. who's D. whose
- The pencil..... you gave me yesterday is lost. A. whose
B. whom C. that D. who
- I honestly don't understand you hate that man. A.
When B. Where C. why D. how
- I'd like to know..... hell wriggle out of this ease A. how
B. whenever C. where D. what
- of the two pencils is yours? A. where B. who C.
which D. whose
- university is..... Institution of higher learning.
A. An... an B. An ... a C. A..... a D. A... an
- I'd give up smoking, if..... you, A. am B. were C. was D.
are
- It is high time you..... visiting that friends of yours.
A. should stop B. stopped C. stop D. must
- had they entered when it started to rain A. hardly
B. nearly C. almost D. fairly
- All your like you very much. A. brother-in-law B.
brother-in-laws C. brother-in-laws D. brothers-in-law

SOLUTION TO LUTH SCHOOL OF NURSING 2015 EXAM

1. C	2. C	3. D	4. A	5. D	6. D	7. E	8. B	9. C	10. D
11. C	12. C	13. C	14. A	15. B	16. E	17. C	18. B	19. B	20. D
21. C	22. B	23. A	24. E	25. B	26. C	27. C	28. D	29. D	30. D
31. D	32. A	33. *	34. B	35. C	36. C	37. D	38. B	39. C	40. B
41. E	42. B	43. C	44. A	45. C	46. B	47. C	48. C	49. C	50. D

1. D	2. D	3. C	4. B	5. B	6. B	7. D	8. C	9. C	10. B
11. A		12. C, B, D, A		15. C	16. C	17. B	18. B	19. A	20. C

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2014

TIME:11%Hour

INSTRUCTIONS TO CANDIDATES

- *Answer all question
- *Affix 1 passport photograph to the answer sheet
- *Write your registration number UME score &state of origin only on the answer sheet
- *Calculators,cell phones and other electronics devices are not allowed.
- *Any form of malpractice automatically disqualifies the candidate
- *Ensure that you sign the attendance sheet before leaving the examination hall.

ENGLISH LANGUAGE

Fill the blank spaces making use of the best of the five options.

- In the past his father used to walk but nowadays heto work by bicycle A.giving B.is going C.seldomly goes D.goes E.has gone
-It might not look such a rosy proposition A.from his viewing proposal B.when he views the proposal C.having viewed the proposal D.from his paint of view E. from his seeing it
- Nuhu told me to go antill seven A.to wark B. worked C.works D.working E.work
- Don't study on the examination day What find he tell you? He told me.....on the examination day A.no studying B don't study C.not study D.not to study E.to not study
- I start writing this test at 8.00am.it is 1000am now by 11.00 am].....it for three hours.A.shall be writing B. have been writing C.shall have been writing D.am writing E.was writing
- I shall travel to Lagos by.....next week A.air or with train B.aircraft or by road C.airways or by bus D.air or by road E.flight or by bus
- Musa asked Asmart what shesince he last saw her A.has been doing B.had been doing C.has done D.was doing E.might be doing
- The evidence of all the acaused person's.....by the judge sitting at no.2 assizes last week A.were disbelieved B.was disbelieved C.in disbelieved D.are disbelieved E.has been disbelieved.

Choose the word or phrase from options A-E which has the same meaning to the underlined word or words in each sentence.

- The minister bit on aplan to lain his past after many months of lobbying A.beat a plan B.drew up a plan C.discovered a plan D.selected a plan E.designed a plan
- Ekwasi's account with the naira hank is in the red,his account A.was written in red ink B.is overdrawn C.is in danger D.is special E.is heavy
- He should heahle to da it alane A.he ought to be able to do it alone B.he may be able to do it alone C.he may be

able to do it alone D.he has been able to do it alone E. he will do it alone

- you can talk to her A.you are permitted to talk to her B. you must talk toher C.you mustn't talk to her D.nothing prevents you from talking to her E.A&D
- When you go to a foreign country to study,you will discover that life is not always a bed of roses after it is A.as pleasant as one thought B.a bed without roses C.an unmitigated d sappointment D.expedted E. uncomfortable
- Ebunthe edge of the cliff after his shoes had called to grip A.fel down B.fell off C.fell away D.fell from
- The host insisted on.....what he called a lttle gift A.me to accept B.me accepting C.my accepting D.my acceptance
- I know you think lam talking nonsense,Shehu,but you will realize that I was right A.at one time B.on time C.in time D.all times
- The driver wll all the students interested to going to lagos tomorrow at 7.00am outside the main building A.lift B.lift up C.pick D.pick up

Choose the option similar in meaning to the underlined word(s)

- The govmment is making concerted efforts at providing essential social facilities in the rural areas A.dissipated B. unconcerted C.unsuccessful D.uncontrolled
- Wecannot but talk about his invaluable contributions to the affaire of the society A.worthless B.cost C.unrecognized D. incalculable
- Men living in an extended family know that they have to be responsible for the ware fare of some other person's children A.externals B.nuclear C.neutral D.contracted

CHEMISTRY

- In the reaction between sodium hydroxide and sulphuric adid solutions,what volume of 0.5molar sodium hydroxide would exactly neutralize 10cm³of 1.25molar sulphuric acid? A.5cm³B.10cm³C.20cm³D.25cm³E.50cm³

22. A small quantity of solid ammonium chloride (NH_4Cl) was heated gently in a test tube. The solid gradually disappears to produce a mixture of two gases later a white cloudy deposit was observed on the cooler part of the test tube. The ammonium chloride is said to have undergone A. distillation B. sublimation C. precipitation D. evaporation E. decomposition
23. Elements P, Q, R, S have 6, 11, 15, & 17 electrons respectively. Therefore A. P will form an electrovalent bond with R B. Q will form a covalent bond with S C. R will form an electrovalent bond with S D. Q will form an electrovalent bond with S E. Q will form a covalent bond with R
24. An element X forms the following compounds with chlorine: NCl_4 , XCl_3 , XCl_2 . This illustrates the A. law of multiple proportions B. law of chemical proportions C. law of simple proportion D. law of conservation of mass E. law of definite proportion.
25. The oxidation state of chlorine in potassium chlorate is A. 11 B. 12 C. 13 D. 15 E. 17
26. When carbon dioxide is bubbled into lime water, a white precipitate is formed. If the passage of the gas is continued the precipitate disappears. The reasons for this is A. calcium carbonate is formed which on reaction dissolves B. calcium hydrogen carbonate is precipitated and then dissolves C. calcium carbonate is formed which on reaction with further carbon dioxide forms soluble calcium hydrogen carbonate D. concentration of solution has occurred with the deposition of calcium hydroxide E. the solution has become saturated and solid carbon dioxide has been deposited.
27. The following reactions are stages in important industrial processes (i) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ ΔH is negative (ii) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ ΔH is negative (iii) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ ΔH is positive. Which of the above forward reactions is favoured? by (i) a decrease in the concentration of the pressure and (ii) an increase in temperature? A. i B. ii C. iii D. i & ii E. i & iii
28. Methanoic acid mixes with water in all proportions and has about the same boiling point as water. Which of the following methods would you adopt to obtain pure water from a mixture of sand, water and methanoic acid A. fractional distillation B. filtration followed by azeotropic distillation C. neutralization with sodium hydroxide followed by distillation D. neutralization with sodium hydroxide followed by filtration E. esterification with ethanol followed by distillation.
29. Which of the following statements applies during the electrolysis of sodium hydroxide solution using platinum electrodes? A. anions are discharged at the cathode B. hydrogen ions are discharged at the anode C. the concentration of sodium hydroxide decreases at both electrode compartments D. the concentration of sodium hydroxide increases at the cathode only E. the concentration of sodium hydroxide increases at the anode only.
30. Which of the following statement is true? When the potassium atom forms its ion A. it gains one electron and becomes neutral B. its atomic number decreases C. it achieves electronic configuration of argon D. it loses one proton E. it loses one neutron.
31. On heating under suitable conditions, 1 liter of a monoatomic gas X combines with 1.5 litres of oxygen forming an oxide. What is the formula of the oxide? A. X_2O_3 B. X_2O C. X_3O_2 D. X_2O_5 E. none of the above
32. If 1 liter of 2M sulphuric acid is poured into a bucket containing 10 liters of water, and the resulting solution is 25 cm³ of 0.1 M Na_2CO_3 , to neutralize it. What is the original volume of the acid? A. 2.2M B. 1.1M C. 0.22M D. 0.1M E. 0.20M
33. A solution of HCl contains 7.3 g dm³ of the acid. What volume of it will neutralize 0.25M solution of Na_2CO_3 of volume 20 ml A. 250 ml B. 50 ml C. 750 ml D. 10 ml E. 25 ml
34. Which one of the following reactions is correct as written? A. $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$ B. $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$ C. $\text{Pb} + 2\text{HCl} \rightarrow \text{PbCl}_2 + \text{H}_2$ D. $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{H}_2\text{O}$ E. $2\text{NH}_4\text{NO}_3 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
35. How many grams of hydrogen gas will be liberated when 12 g of magnesium ribbon dissolves in 500 cm³ of 6M HCl? (Mg = 24, H = 1, Cl = 35.5). A. 2.2 g B. 0.7 g C. 0.5 g D. 0.3 g E. 12 g
36. An example of a strong electrolyte is A. ethyl acetate B. ethanol C. glucose D. sodium formate E. formic acid
37. The oxidation state of manganese in KMnO_4 is A. 44 B. 13 C. 15 D. 17 E. 12
38. For iron to rust, these should be present A. oxygen B. moisture C. carbon dioxide D. oxygen and moisture E. oxygen, moisture and carbon dioxide.
39. The mass number of an atom is A. the mass number of electrons in the atom B. the number of protons in the nucleus of the atom.
40. Sodium and potassium belong to the same group of the periodic table. This is because A. both have identical electronic configuration B. both react with water vigorously.

MATHEMATICS

41. After getting an increase of 15%, a man's new monthly salary is N345. How much per month did he earn before the increase? A. N3308 B. 396.75 C. N3000 D. N393.25 E. N360
42. In base ten, the number 101101 (base 2) equals A. 2010 B. 4510 C. 1510 D. 1310
43. The annual profits of a transport business were divided between the two partners A and B in the ratio 3:5. If B received N3000 more than A, the total profit was A. N5000 B. N18000 C. N12000 D. N24000 E. N8000.
44. A square of cardboard is taped at the perimeter by a piece of ribbon 20 cm long. What is the area of the board? A. 20 sq. cm B. 25 sq. cm C. 36 sq. cm D. 100 sq. cm E. 16 sq. cm
45. Simplify $62^{\frac{1}{2}} \times 16^{\frac{1}{4}} \times 5^{\frac{1}{2}} \times 5^{\frac{1}{4}}$
46. The median of the set of numbers 4, 94, 13, 7, 1410, 17 is A. 13 B. 7 C. 3 D. 10 E. 7
47. List all integer values of x satisfying the inequality: $1 < 2x - 5 \leq 5$ A. 2, 3, 4, 5 B. 2, 5 C. 3, 4, 5 D. 2, 3, 4 E. 3, 1.
48. Find the roots of the equation $10x^2 - 13x - 3 = 0$. $x = \frac{3}{8}$ or $-\frac{1}{2}$ B. $x = \frac{3}{10}$ or $-\frac{1}{5}$ C. $x = \frac{3}{10}$ or $\frac{1}{5}$ D. $-\frac{1}{5}$ or $\frac{3}{2}$
49. A solid cylinder of radius 3 cm has a total surface area of



- 366cm². find its height. A. 2cm B. 3cm C. 4cm D. 5cm
50. If $\sin^2 \theta = \frac{e}{b}$, what is $\sin(90^\circ - \theta)$? A. $\frac{a}{b}$ B. $1 - \frac{a}{b}$ C. $\frac{b}{a}$ D. $1 - \frac{b}{a}$
51. 7 pupils of average age 12 years leave a class of 25 pupils of average age 14 years. If 10 new pupils of average age 11 years join the class, what is the average age of the pupils now in the class? A. 13 years B. 12 years C. 13 years 5 months D. 13 years E. 11 years
52. A sum of money invested at 5% per annum simple interest amounts to \$434.00 at 7 1/2% per annum simple interest? A. 7 years B. 10 years C. 5 years D. 12 years E. 14 years.
53. By selling an article for N45.00 a man makes a profit of 8%. For how much should he have sold it in order to make a profit of 32%? A. N180.00 B. N59.00 C. N63.00 D. N58.00 E. N55.00
54. An isosceles triangle of sides 13cm, 13cm, 10cm is inscribed in a circle. What is the radius of the circle? A. 7.1124cm B. 12cm C. 5cm D. 7cm E. 69cm
55. A man bought a wristwatch for N150 but was only able to sell it for N120. Find the loss per cent on the transaction. A. 25% B. 11 1/8% C. 20% D. 80% E. 30%
56. Find the median of the set of numbers 110, 116, 113, 118, 121, 118, 117, 113. A. 117.5 B. 118 C. 117 D. 116 E. 113.
57. Given a regular hexagon, calculate each interior angle of the hexagon. A. 600 B. 30°C. 120° D. 45° E. 135°
58. Solve the following equations: $4x - 3 = 3x + y$ $2y + 5x = 12$; A. $x = 5, y = 2$ B. $x = 2, y = 5$ C. $x = 2, y = -5$ D. $x = 5, y = 3$
59. A man has a certain number of mangoes. He gives 30% to his son, 40% to his daughter, and 30% to his wife. He is left with 16 mangoes. How many mangoes did he have at first? A. 36 B. 26 C. 36 D. 16 E. 36
60. Which of the following lines is parallel to the line $3y + 2x - 17 = 0$? A. $3y + 2x + 7 = 0$ B. $9y + 6x + 17 = 0$ C. $24y + 16x + 19 = 0$ D. $3y - 2x + 70 = 0$ E. $15y + 10x - 10x - 13 = 0$
- PHYSICS
61. A piece of rubber 10cm long stretches 6mm when a load of 100N is hung from it. What is the area stretched if the young modulus is 50N/m². A. 60m² B. 150m² C. 33.33m² D. 15m²
62. To determine the weight of an object you would A. use a balance B. use a spring balance C. find the force necessary to give it a certain acceleration D. use none of these methods E. use any of these methods.
63. A gas at pressure PN/m and temperature 27°C is heated to 77°C at constant volume. The new pressure is A. 0.85PN/m B. 0.86PN/m C. 1.16PN/m D. 1.18PN/m E. 2.85PN/m
64. Two lamps rated 40W and 220W, each are connected in series. The total power dissipated in both lamps is A. low B. 20W C. 40W D. 80W E. none
65. A magnet is moved through a coil of wire. The emf produced in the wire depends on A. the number of turns in the coil B. the strength of the magnet C. the speed at which the magnet is moved D. all of the above E. none of the above
66. A potential difference of 10V is used to produce a current of 2A for 200s through a heating coil. The heat produced is A. 4800cal B. 6000cal C. 2400J D. 240kcal E. 600J
67. Two boys are communicating with each other by stretching a string passing through a hole punched in the bottom of each of the two tin cans. The physical principle employed is that

sound travels A. mainly through air B. fainter in stretched string C. faster through gases than in solids and liquids D. with greater ease through a string than in air E. none of the above is correct.

68. The hatch door of a submarine has an area of 0.5m² the specific gravity of sea water is 1.03. Assume that $g = 10\text{m/s}^2$ and neglect the atmospheric pressure. The force exerted by the sea water on the hatch door at a depth of 200m is A. $1.03 \times 10^6\text{N}$ B. $1.03 \times 10^5\text{Nm}^2$ C. $26 \times 10^3\text{Nm}^2$ D. 2.06 x 10¹¹N E. $1.03 \times 10^2\text{N}$.
69. When equal weights of iron and water are subjected to an equal supply of heat, it is found that the piece of iron becomes much hotter than water after a short time because A. The specific heat of iron is higher than water B. Iron is in solid form C. water is in liquid form D. the specific heat of water is higher than that of iron E. the specific heat of iron is infinite.
70. The speed of light in air is $3.0 \times 10^8\text{m/s}$. Its speed in glass having a refractive index of 1.65 is A. $1.82 \times 10^8\text{m/s}$ B. $3 \times 10^8\text{m/s}$ C. $6 \times 10^8\text{m/s}$ D. $1.82 \times 10^2\text{m/s}$
71. An electric iron is rated at 100W, 250V. The corresponding maximum resistance and accompanying current is A. 62.5Ω, 4.032A B. 16.00Ω, 6.25A C. $4.95 \times 10^8\text{Ω}$, 4.00A D. 62.5Ω, 4.0A E. 4.0Ω, 250A.
72. 250g of lead at 170°C is dropped into 10g of water at 10°C. If the final temperature is 12°C, the specific heat of lead is (specific heat capacity of water is 4,200J/kg°C) A. 39.5J/kg°C B. 50.4J/kg°C C. 127.5J/kg°C D. 154.6J/kg°C E. 173.4J/kg°C.
73. A positively charged rod is brought nearer to the top cap of a positively charged gold leaf electroscope. The divergence of the leaves will A. decrease to zero B. steadily increase C. decrease to zero and then increase D. remain constant E. increase to a maximum and then decrease.
74. A rocket burns 0.01kg of fuel each second and ejects it as a gas with velocity of 5,000m/s. What force does the gas exert on the rocket? A. 500,000N B. 500N C. 50N D. 5,000N E. 50,000N.
75. Which of the following assumptions is made in a simple pendulum experiment? The A. suspending string is inextensible B. bob has a finite size C. bob has a definite mass D. initial angle of oscillation must be large.
76. A rectangular tank contains water to a depth of 2m. If the base is 4m x 3m, calculate the force on the base (density of water 10^3kg/m^3 , $g = 10\text{m/s}^2$) A. $24 \times 10^3\text{N}$ B. $24 \times 10^4\text{N}$ C. $2.0 \times 10^4\text{N}$ D. $1.7 \times 10^4\text{N}$
77. The expansion of solids can be considered as disadvantage in the A. balance wheel of water B. fitting of wheels on our vehicle C. fire alarm D. thermostat.
78. When two objects P and Q are supplied with the same quantity of heat, the temperature change in P is observed to be twice that of Q. The ratio of the specific heat capacity of P to Q is A. 1:4 B. 4:1 C. 1:1 D. 2:1
79. A lens of focal length 12.0cm forms an upright image three times the size of a real object. The distance

between the object and the images is A.8.0cm B.16.0cm
C.24.0cm

D.32.0cm

BIOLOGY.

81. Anatomy is the study ofand A. forms and structure B. forms and cell C. Tissue and organ D. system and structure E. A &
82. is the systematic study of the natural phenomena in the environment A. Anatomy B. cell C. Science D. botany E. polymer
83. Cell comes from pre-existing cell was a law given by A. Charles Darwin B. Charles Lams C. Robert Hooke D. Robert Korch E. A&B
84. Using the punnet square given by Whitaker, the genotypic ratio in the second law is A. 3:1 B. 1:3:1 C. 1:2:1 D. 2:1 E. 4:3:1
85. Groups of cell grouped together link up to produce A. Cells B. Tissue C. Organ D. system E. AEB.
86. The Capacity of living organism to respond to internal and external stimuli is A. Excretion B. nutrition C. irritability D. respiration
87. Organism possesses living and non living quality A. Viruses B. Fungi C. Algae D. A&B E. A&C
88. Carl Linnaeus proposed System of naming A. 10 B. 5 C. 6 D. 7 E. 12
89. The role of organism in the environment is termed A. Biome B. Niche C. Ecosystem D. Catalyst E. A&C
90. Protoplasm + Nucleus + + +
91. The outer covering of plant cell is A. membrane B. cell wall C. cytoplasm D. plasma
92. The heart pumps blood to all part of the body True/False
93. Blood vessel carries blood away from the heart. A vein B. capillaries C. arteries D. Nodules
94. The shoulder blade is also called A. Scalp B. Tissue C. Scapular D. Tendon E. Ligament
95. Ribosomes is one of the organelle responsible for A. fighting foreign bodies B. synthesis of protein C. building offats D. disintegrate lipids.
96. The study of how living organism interact in its environment is termed A. ecology B. ecosystem C. Niche D. Zoology
97. Mendel's law on genetics is termed as the law of A. Selection B. Assortment C. Segregation D. Build up E. A&C
98. Pairs of gene with alternating form is termed A. Pure line B. Chromosome C. Allele D. Genes
99. Strands of genetic material are termed as A. Pure line B. Chromosome C. Autosome D. Manosomes
100. is not a vertebrae A. Cervical vertebrae B. Caudal vertebrae C. Kosal vertebrae D. Sacral vertebrae.

SOLUTION TO 2014 LUTH SCHOOL OF NURSING ADMISSIONS SELECTION TEST

1. D	2. A	3. D	4. D	5. A	6. D	7. D	8. B	9. C	10. C
11. C	12. E	13. A	14. D	15. D	16. C	17. A	18. A	19. A	20. C
21. E	22. B	23. D	24. A	25. C	26. C	27. C	28. B	29. B	30. C
31. B	32. C	33. B	34. D	35. C	36. E	37. D	38. D	39. E	40. A
41. C	42. B	43. C	44. B	45. C	46. A	47. C	48. D	49. C	50. C
51. C	52. B	53. B	54. E	55. C	56. C	57. C	58. B	59. C	60. B
61. B	62. B	63. C	64. E	65. A	66. D	67. D	68. E	69. A	70. A
71. A	72. C	73. A	74. B	75. C	76. A	77. A	78. B	79. D	80. A
81. A	82. C	83. C	84. C	85. B	86. C	87. A	88. B	89. B	90. C
91. B	92. TRUE	93. C	94. C	95. B	96. B	97. C	98. C	99. B	100. C

We wish you success!